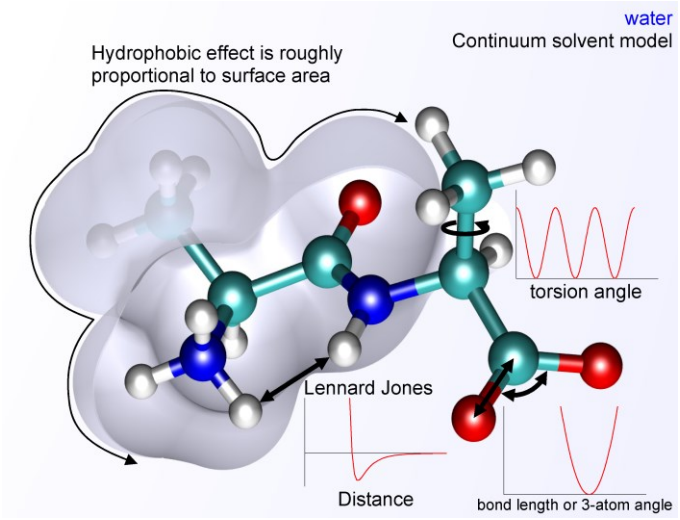
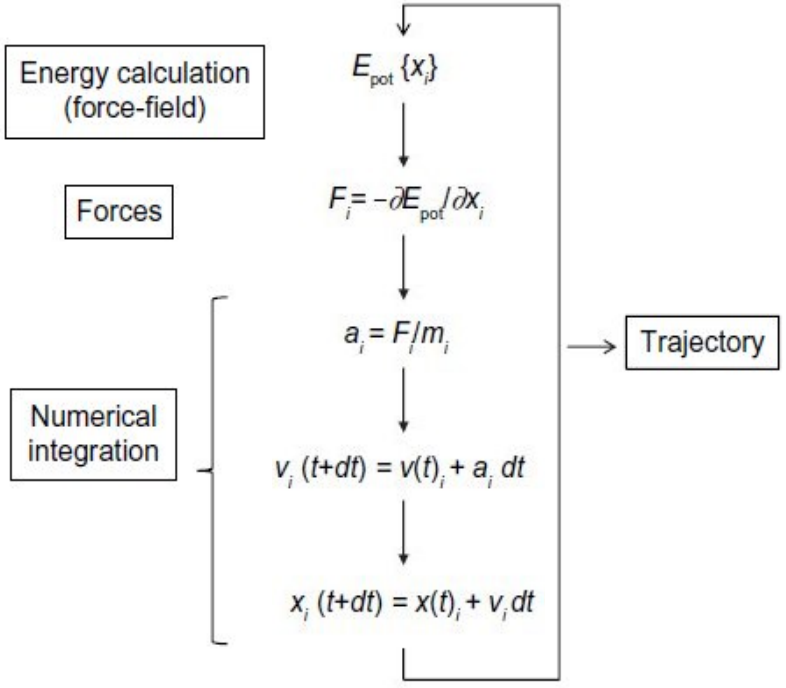
Methodology Simplified introduction to molecular modelling: bridging the gap

Assuming every particles in a system behave according to the *Newton's 2nd law of motion:*

$$F=ma$$





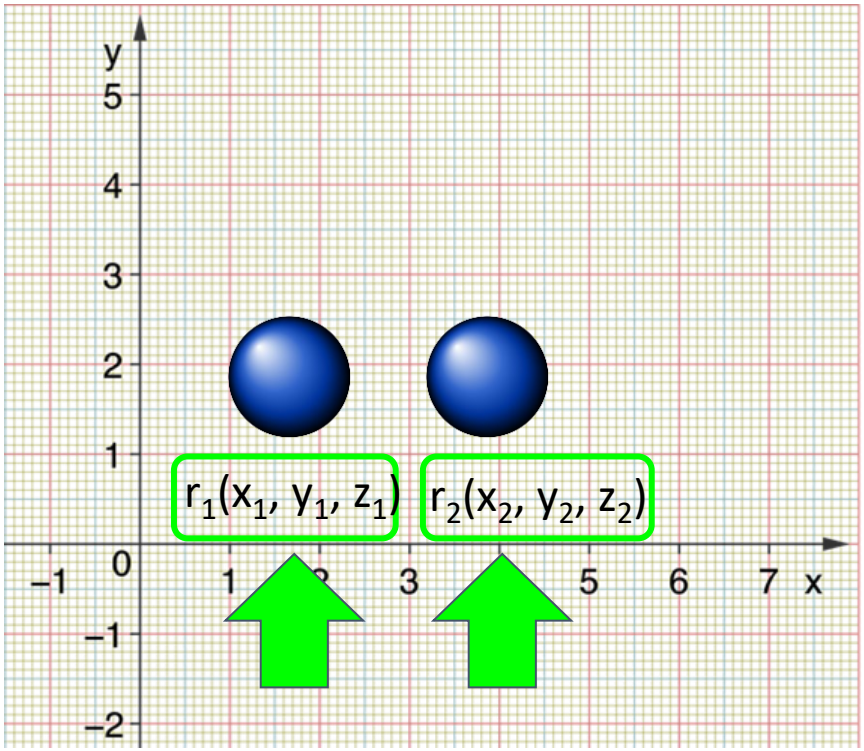
Predict the properties of "**unknown**" based on limited data available:

- Composition
- Structural data

Image source:

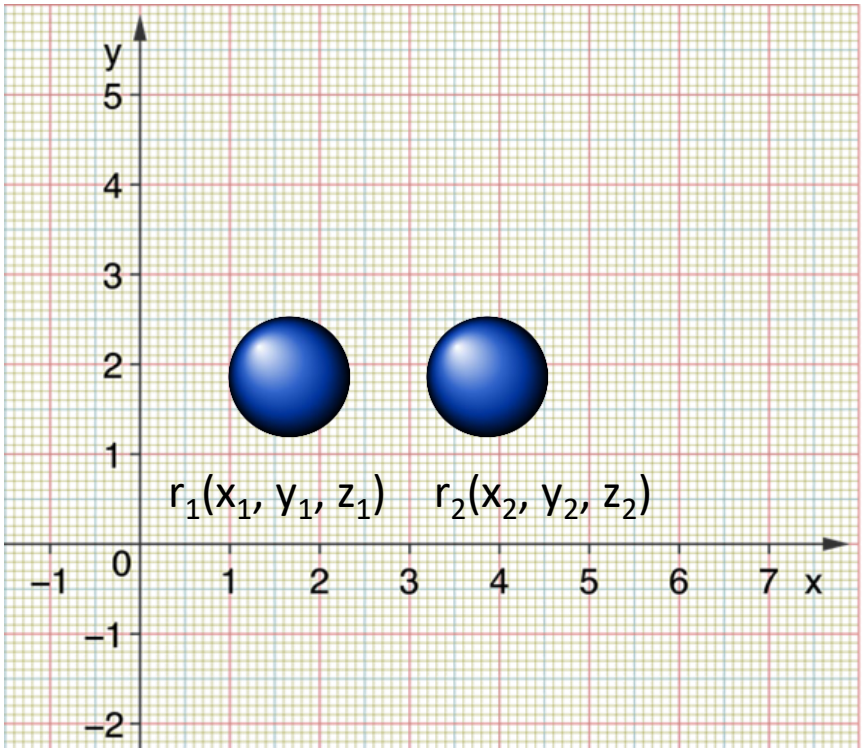
1. https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Advanced_Theoretical_Chemistry_(Simons)/07%3A_Statistical_Mechanics/7.03%3A_Molecular_Dynamics_Simulations
2. https://www.dovepress.com/molecular-dynamics-simulations-advances-and-applications-peer-reviewed-fulltext-article-AABC
3. Image adapted from Wikimedia commons.

Methodology Simplified introduction to MD simulation

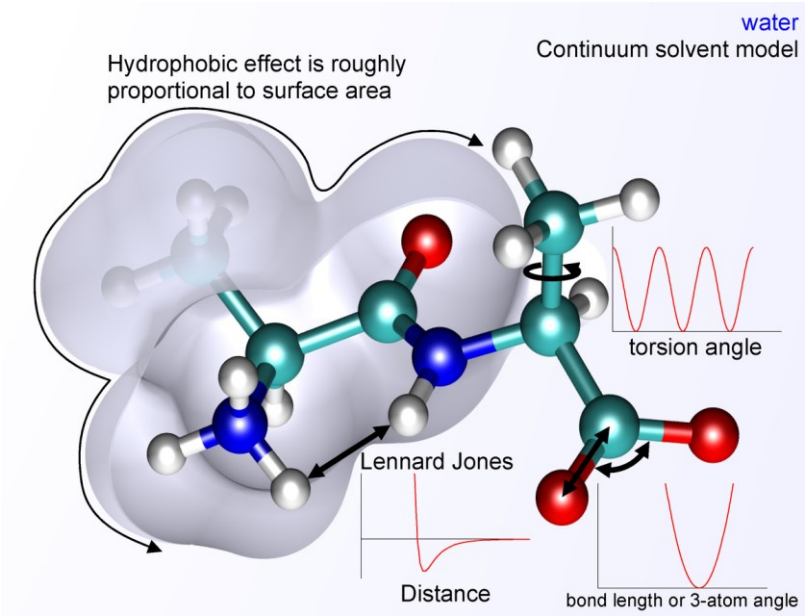


- Input initial conditions:
  - Position: r

Methodology Simplified introduction to MD simulation

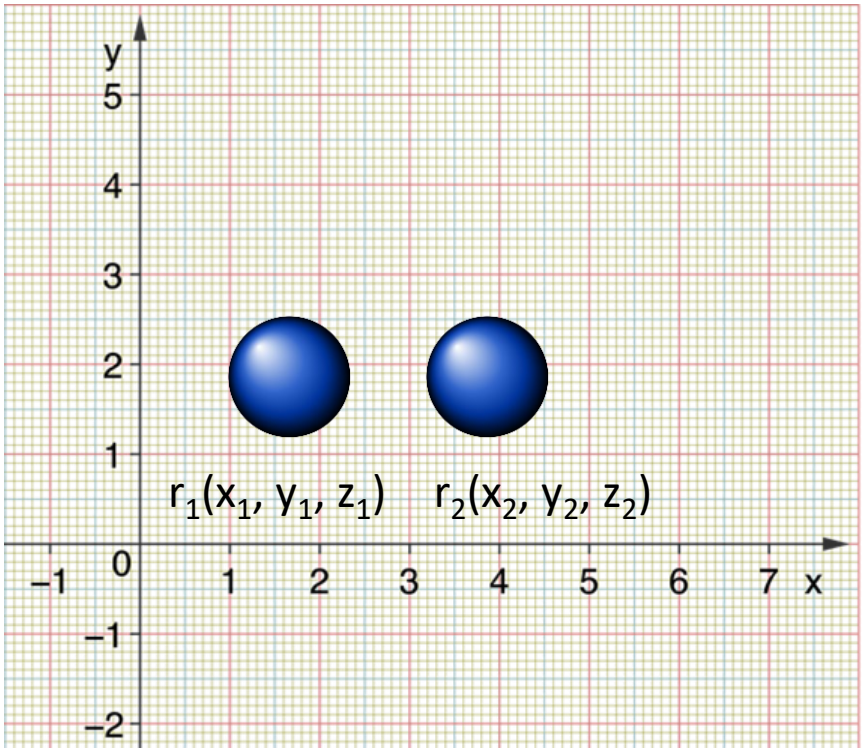


- Input initial conditions:
  - Position: r
  - Potential interaction: V (Force field)



https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Advanced_Theoretical_Chemistry_(Simons)/07%3A_Statistical_Mechanics/7.03%3A_Molecular_Dynamics_Simulations

Methodology Simplified introduction to MD simulation



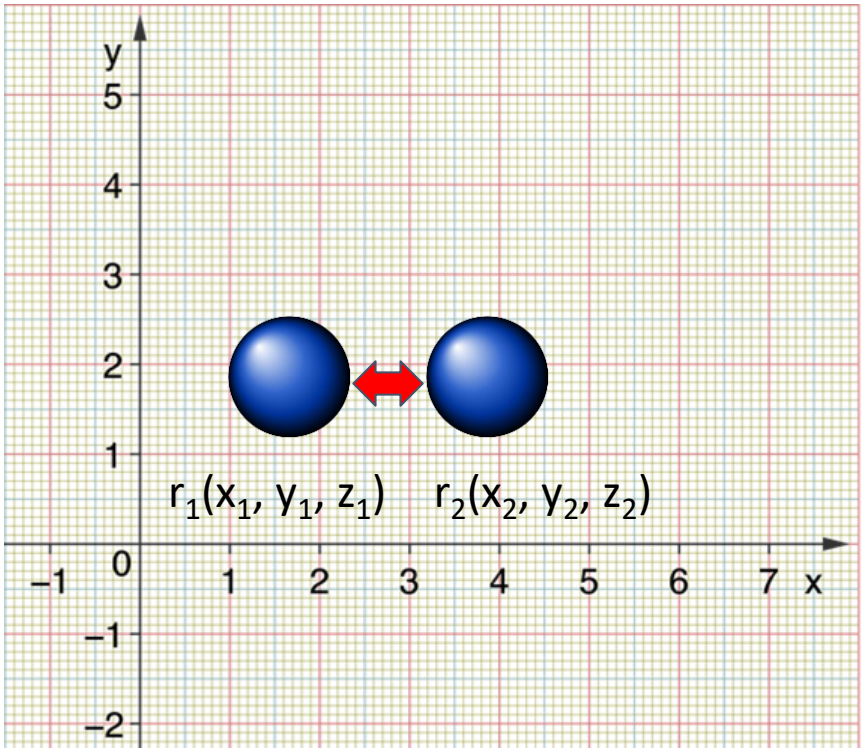
- Input initial conditions:
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$$E_{total} = E_{bonds} + E_{angles} + E_{dihedrals} + E_{vdw+electrostatic}$$

$$= \sum_{bonds} K_r(r - r_{eq})^2 + \sum_{angles} K_\theta(\theta - \theta_{eq})^2 +$$

$$\sum_{dihedrals} \frac{V_n}{2}[1 + \cos(n\phi - \gamma)] + \sum_{i<j} [\frac{A_{ij}}{R_{ij}^{12}} + \frac{B_{ij}}{R_{ij}^{6}} + \frac{q_i q_j}{\epsilon R_{ij}}]$$

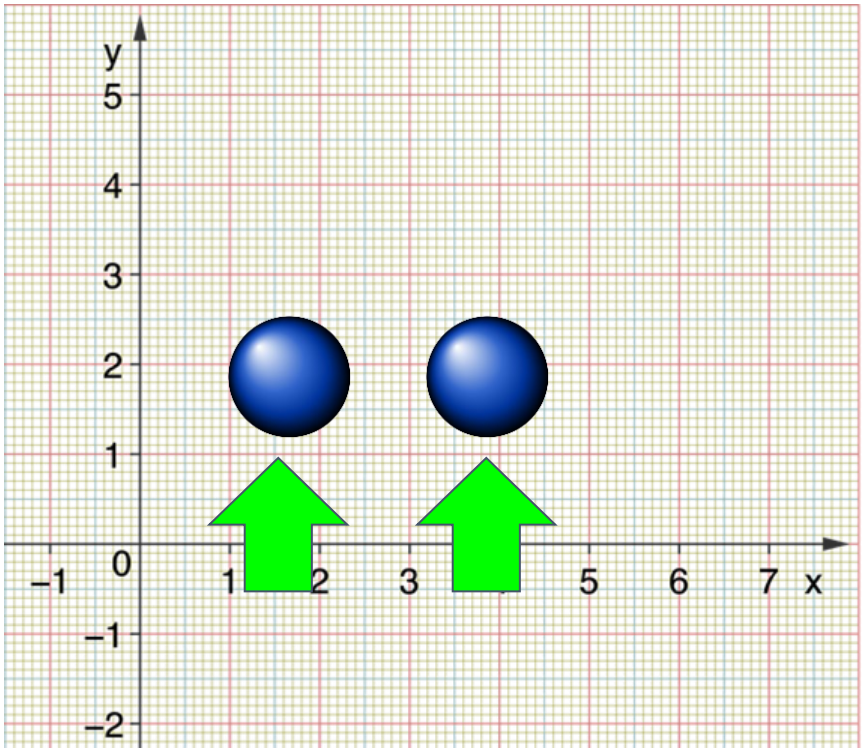
Methodology Simplified introduction to MD simulation



- Input initial conditions:
  - Position: r
  - Potential interaction: V (Force field)
- Calculate forces: Newton's 2nd law

$$F = ma = m\frac{\partial^2 r}{\partial t^2} = -\frac{\partial V}{\partial r}$$

Methodology Simplified introduction to MD simulation

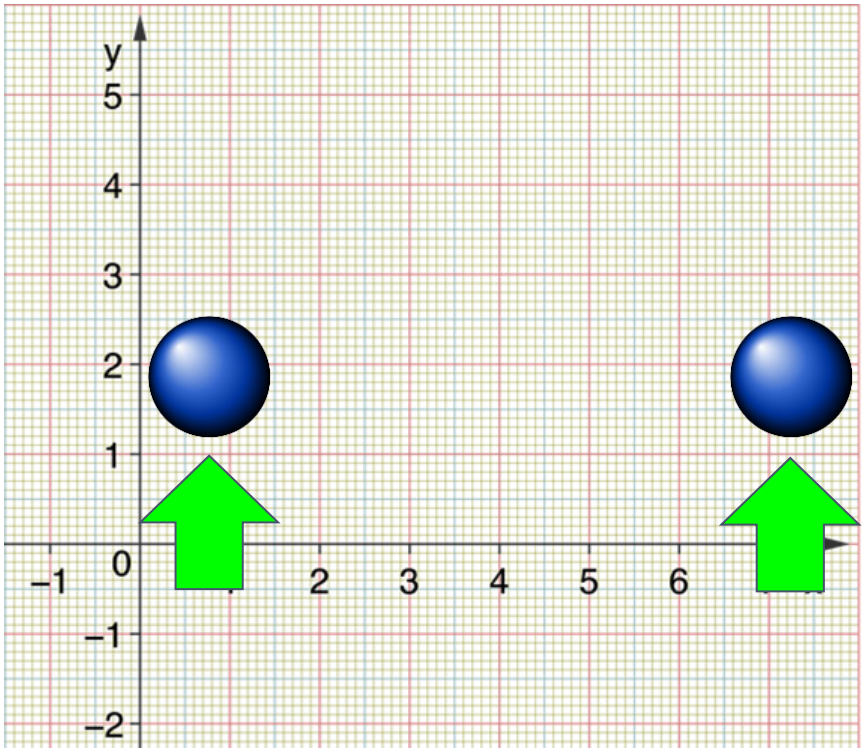


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- Update configurations

Methodology Simplified introduction to MD simulation



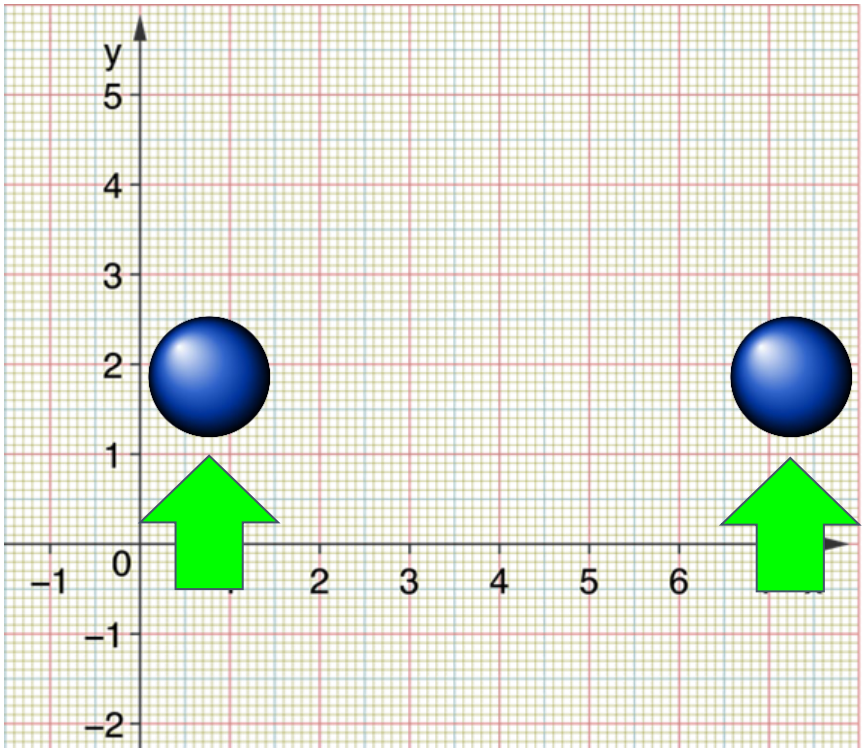
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$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = \frac{F}{m}$$

Methodology Simplified introduction to MD simulation



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Predict the properties of "**unknown**" based on limited data available:

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Methodology Simplified introduction to MD simulation

